Research questions from the current analysis

Multimodal drivers + TerraMind embeddings for daily soil moisture

Prajwal Khanal

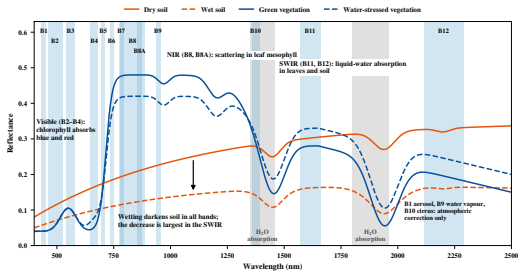
7 October 2026

Motivation Different ways of obtaining soil moisture

Source	How SM is obtained	Depth	Main limitation
Microwave (SMAP, SMOS, ASCAT, S1)	invert: σ^0 or $T_B = f(\mathrm{SM}, \mathrm{roughness}, \mathrm{vegetation})$	top $\sim$5 cm; informs part of the root zone (Feldman et al., 2023)	shallow; assumes vegetation and roughness; coarse (9–40 km) except S1
Optical / thermal (S2, Landsat, MODIS)	empirical relations with surface properties: reflectance and indices; LST–vegetation space (e.g. triangle method, TVDI)	skin / surface	indirect, site-specific calibration; clouds; canopy hides the soil
Land-surface models	$d\theta/dt = P - ET - Q - D$, with parameters p	full profile	forcing errors, soil and vegetation parameters, process representation (Koster et al., 2009; Seneviratne et al., 2010)
Reanalysis (ERA5-Land)	LSM driven by reanalysis meteorology	full profile	model assumptions dominate deep layers with few observational constraints (Li et al., 2021; Miguez-Macho and Fan, 2021)
In situ (ISMN)	direct probe measurement	several depths	point scale; sparse, uneven global coverage
Data-driven (ML / DL)	$\mathrm{SM} = f_\theta(\mathrm{inputs})$, learned from in-situ labels; no ad-hoc process assumptions	any depth with labels	needs enough, representative sites; generalisation?; hard to interpret?

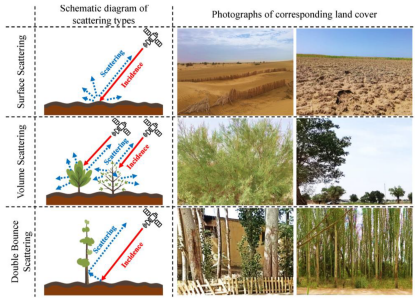
Motivation Multimodal remote sensing

1. Optical: reflectance spectrum



- Dry and wet soil reflect differently: wetting darkens soil in every band, most in the SWIR (B11, B12)
- Green and water-stressed vegetation reflect differently: stress lowers NIR and raises SWIR

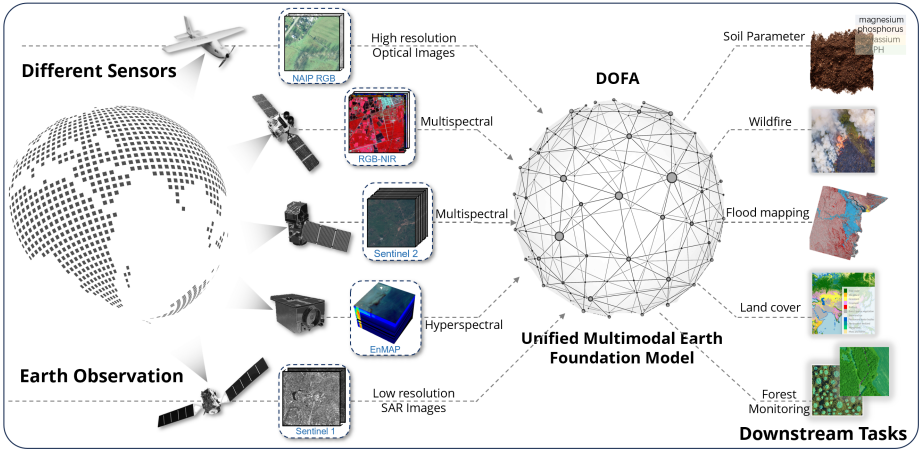
2. SAR: scattering mechanisms



Nurmemet et al. (2025), *Sensors* 25(8), 2512, doi:10.3390/s25082512

- Bare soil: surface scattering, set by moisture (dielectric) and roughness
- Canopy: volume and double-bounce scattering mask the soil signal

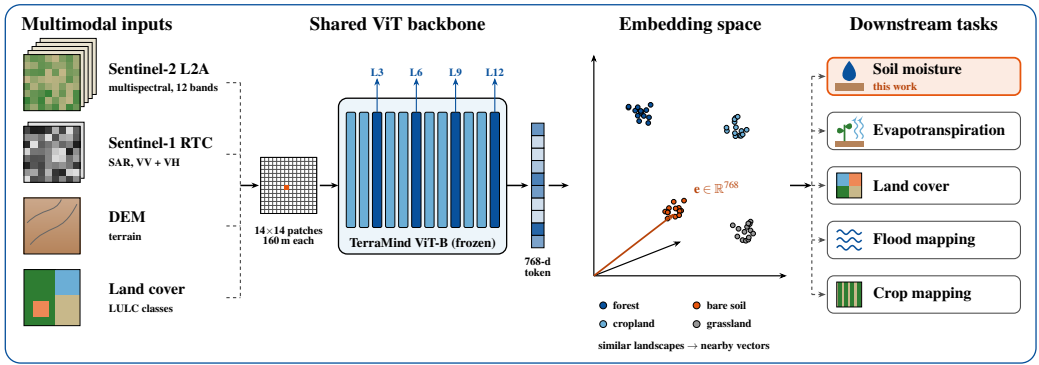
Motivation Geospatial foundation models (GFMs)



DOFA: Xiong et al. (2024), Neural plasticity-inspired multimodal foundation model for Earth observation, arXiv:2403.15356

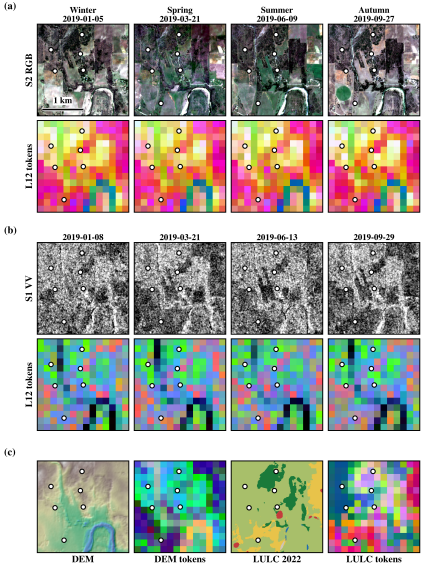
Once pre-trained on multimodal, unlabelled data, its representation works for many downstream tasks

Motivation TerraMind embeddings for soil moisture



Frozen TerraMind ViT-B turns each 160 m patch of S2, S1, DEM and LULC into a 768-d embedding; the same embedding can serve many tasks; in this case, **soil moisture**.

Motivation Example: TerraMind embeddings at one site (TxSON, 2019)



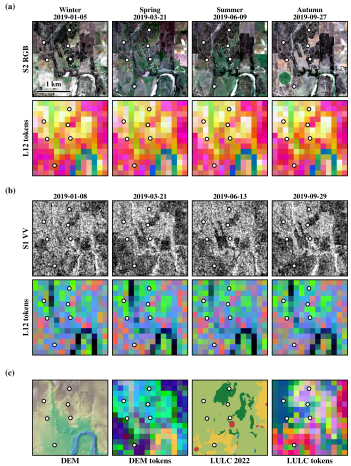
S2, S1, DEM and LULC with their TerraMind L12 tokens.

Each 160 m patch → one 768-d vector; colours show its first 3 principal components (one basis per row, each panel centred on its own mean).

Eight research questions

Block	RQ	Question
Core	RQ1	Do frozen GFM embeddings add skill beyond a multimodal driver baseline?
	RQ2	What does the model generalise to — places or processes?
	RQ3	How does skill vary across the landscape?
Mechanism	RQ4	Which modalities matter, at which depth?
	RQ5	Does the fine-resolution path carry real sub-km information?
Negative	RQ6	Is land-surface temperature a useful auxiliary signal for SM?
Outlook	RQ7	Can the model detect management signals (irrigation)?
	RQ8	Does the representation transfer to an unseen region?

RQ1 Do frozen GFM embeddings add skill beyond a multimodal driver baseline?



TerraMind L12 tokens for S2, S1, DEM and LULC at one tile (TxSON, 2019).

Have

- Modality-shuffle ablation, §59 model, 221 OOS stations (next slide)
- Satellite tokens swapped across stations: +37 % ubRMSE, r 0.77 $\rightarrow$ 0.52

Missing

- Drivers-only model (10 epochs)
- Raw-imagery control (GFM vs pixels)
- Within-station S2 shuffle

RQ1 What does the model use the GFM embeddings for? (shuffle ablation)

	0–10 cm		10–30 cm		30–100 cm	
Shuffled input	ΔubRMSE (r)	worse	ΔubRMSE (r)	worse	ΔubRMSE (r)	worse
All satellite, cross	+36.8% (0.52)	97%	+37.1% (0.52)	96%	+38.4% (0.44)	92%
S2, cross	+4.9% (0.74)	84%	+6.8% (0.72)	87%	+8.7% (0.63)	85%
S1, cross	+4.8% (0.73)	82%	+7.7% (0.70)	84%	+14.4% (0.58)	86%
S1, within	0.0% (0.75)	49%	+1.0% (0.73)	59%	0.0% (0.64)	49%
DEM, cross	+12.6% (0.71)	90%	+9.7% (0.68)	86%	+18.4% (0.59)	96%
LULC, cross	+6.1% (0.73)	82%	+6.6% (0.70)	82%	+12.7% (0.62)	87%
Fine 20 m path, cross	+0.5% (0.75)	56%	+1.3% (0.74)	65%	+0.1% (0.68)	52%
ERA5, cross	+30.3% (0.50)	95%	+24.6% (0.54)	91%	+15.1% (0.50)	86%
ERA5, within	+39.0% (0.35)	88%	+30.4% (0.37)	81%	+13.5% (0.34)	73%
Soil, cross	+7.6% (0.72)	87%	+17.6% (0.63)	92%	+25.7% (0.55)	94%
SIF, cross	+1.3% (0.75)	63%	+1.6% (0.73)	62%	+1.8% (0.67)	71%
TWSA, cross	0.0% (0.77)	50%	0.0% (0.76)	53%	+0.1% (0.71)	58%

Baseline: ubRMSE 0.050 / 0.048 / 0.048, r 0.766 / 0.761 / 0.702 (n = 221 / 144 / 119). *Cross* = another station's input (kills site identity); *within* = same station, shuffled dates (kills timing).

RQ1 What we can and cannot claim today

Can claim

1. Reliance on GFM tokens $\approx$ reliance on weather (+37 % vs +30 %).
2. Tokens are a *static landscape descriptor*: S1 within-station shuffle = 0 %.
3. They shape the response, not just the mean (ubRMSE is bias-free): *GFM embeddings condition the weather$\rightarrow$SM transfer on the landscape.*
4. Depth pattern inverted vs a moisture signal: S1 +4.8 % $\rightarrow$ +14.4 % with depth.
5. Modalities redundant: S2+S1+DEM+LULC $\approx$ 28 % $<$ 37 % joint.
6. Fine 20 m path and TWSA unused.

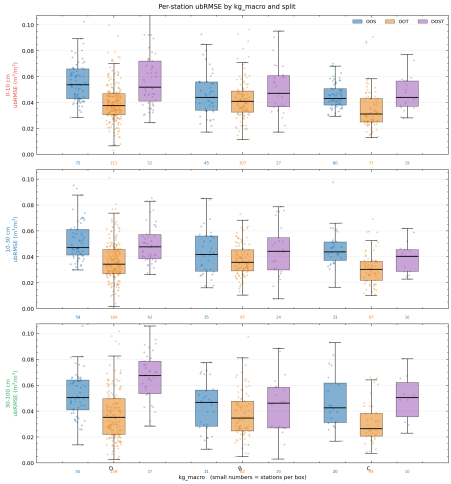
Cannot claim yet

- *Added* skill — a shuffle measures reliance
- GFM better than raw imagery
- Anything about S2 dynamics

To answer fully

- Within-station S2 shuffle (eval only)
- Run A: drivers + soil, no tokens
- Run B: A + static site descriptors
- Full $>$ B $>$ A $\Rightarrow$ dynamic skill;
  Full $\approx$ B $>$ A $\Rightarrow$ site info only

RQ2 What does the model generalise to — places or processes?



Per-station ubRMSE by climate zone and split (OOS / OOT / OOST).

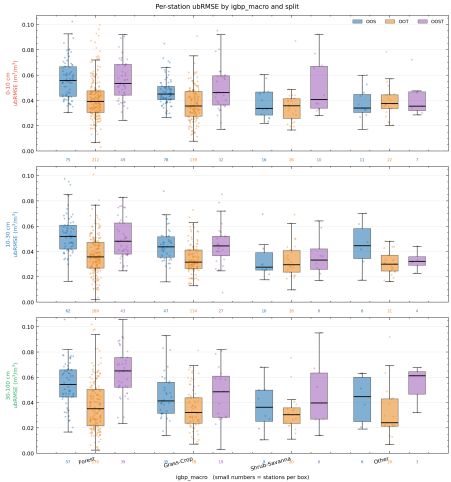
Have

- Splits §47: train 573 / val 89 / OOS 224; OOT 2023–2025
- OOT error is *lower* than OOS in every climate zone: new years at known sites are easy, new sites are hard
- `pw_stage2a_L3`: OOT 0.0434 beat val 0.0507
- Step-0 offsets probe: pair sign 0.50; site-fingerprint verdict confirmed (27572594)

Missing

- One val / OOT / OOS table for the selected model

RQ3 How does skill vary across the landscape?



Per-station ubRMSE by land-cover class and split.

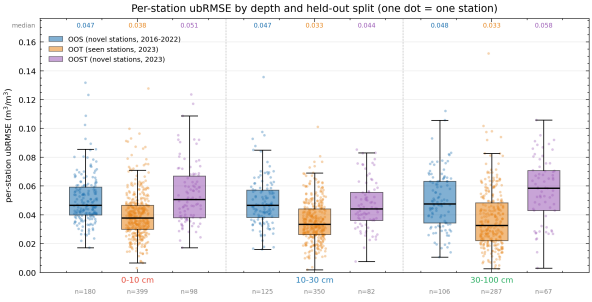
Have

- Per-station metrics by climate zone and land cover
- LULC, elevation band, soil, HAND for every station
- HAND stable under DEM noise; TWI is not — use HAND

Missing

- Error vs aridity, HAND, soil texture, per depth
- Flag elevation sentinels first (18 REMEDHUS = −99.9, Baytik = 0)

RQ4 Which modalities matter, at which depth?



Per-station ubRMSE by depth.

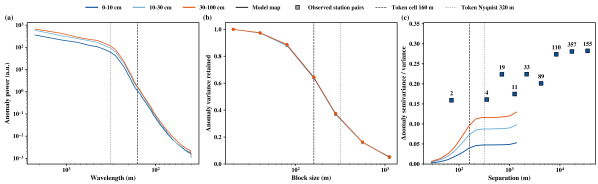
Have (shuffle ablation, RQ1 table)

- ERA5 dynamics dominate the surface (+39 %) and fade with depth (+14 %)
- Soil properties grow with depth (+7.6 % → +25.7 %)
- SIF ≈ 1.5 %; TWSA ≈ 0
- §58: 180-day ERA5 window

Hypothesis status

- "TWSA drives the root zone" — not supported so far

RQ5 Does the fine-resolution path carry real sub-kilometre information?



Effective-resolution test at TxSON (§70).

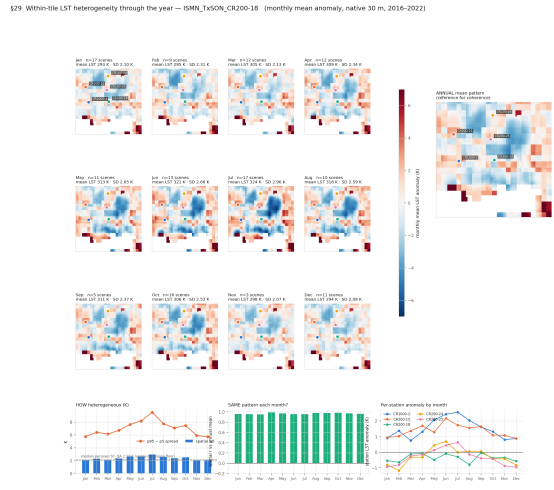
Have

- §53 fine ablation: SM ignores the 20 m path (LST reads it)
- Fine-path shuffle: +0.1 to +1.3 % ubRMSE
- §48 map flattens after epoch 8
- Overijssel grid run (§68, paused)

Framing

- A publishable negative result if framed well

RQ6 Is land-surface temperature a useful auxiliary signal for soil moisture?



Within-tile Landsat LST anomaly by month, TxSON CR200-18 (§29).

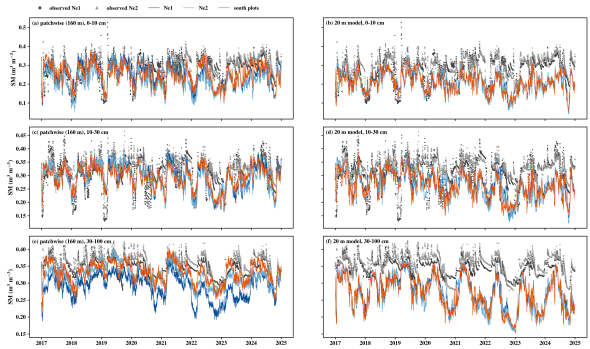
Have (all negative)

- Within-tile LST pattern is static: $r = +0.967$ across months
- Simpson's paradox: +0.167 pooled, −0.077 within station
- ECOSTRESS DTR = day pattern (r = +0.816); G0 fails
- In training LST hurts SM: val 0.0537 with vs 0.0487 without

Status

- Ready to write as a short paper / section

RQ7 Can the model detect management signals such as irrigation?



Mead Ne1/Ne2/Ne3: model vs observed SM, 2017–2024 (§69).

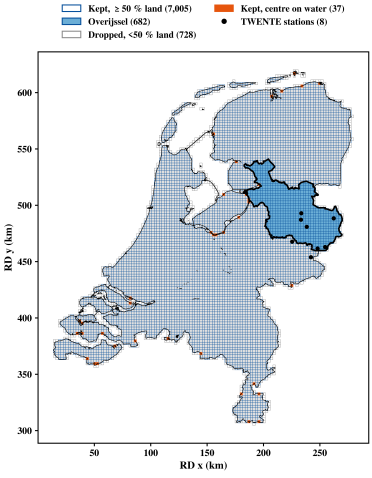
Have

- Mead daily SM built for Ne1/Ne2/Ne3
- Irrigation visible only in the dry year (2022)

Missing

- Irrigation log covers 2014–2021 only
- Prediction-minus-observation analysis vs irrigation dates

RQ8 Does the representation transfer to an unseen region?



Overijssel grid tiles (§68).

Have

- Overijssel: cloud mask done 688/688
- TWENTE: 8 stations, all OOS (end 2020, no OOT)

Missing

- Prep → tokenise → post → predict
- Map + TWENTE station comparison